

8. Haemochromatosis is a genetic condition which causes the body to absorb more iron than normal from the diet. The excess iron is stored in the body's tissues and organs where it can cause damage. The condition mainly affects people of Northern European origin.

Haemochromatosis is caused by a mutation to the HFE gene. The HFE gene provides instructions for producing a protein which regulates iron levels in liver cells. The mutant allele is recessive to the normal allele.

(a) (i) State the meaning of the following terms:

I. gene; [1]

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II. mutation. [1]

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(ii) State the scientific technique which could be used to identify the presence of the mutant HFE allele. [1]

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(b) **Table 8.1** shows the amino acids coded for by each triplet code of DNA.

Table 8.1

		Second letter				Third letter
		T	C	A	G	
First letter	T	TTT } phe TTC } TTA } leu TTG }	TCT } TCC } ser TCA } TCG }	TAT } tyr TAC } TAA stop TAG stop	TGT } cys TGC } TGA stop TGG trp	
	C	CTT } CTC } leu CTA } CTG }	CCT } CCC } pro CCA } CCG }	CAT } his CAC } CAA } gln CAG }	CGT } CGC } arg CGA } CGG }	
	A	ATT } ATC } ile ATA } ATG met	ACT } ACC } thr ACA } ACG }	AAT } asn AAC } AAA } lys AAG }	AGT } ser AGC } AGA } arg AGG }	
	G	GTT } GTC } val GTA } GTG }	GCT } GCC } ala GCA } GCG }	GAT } asp GAC } GAA } glu GAG }	GCT } GGC } gly GGA } GGG }	

The DNA base sequences in **Image 8.2** show part of the normal and mutant HFE alleles found at the same position on a pair of chromosomes and part of the order of amino acids for which they code.

- (i) Use the information provided in **Table 8.1** to insert the missing amino acids in **Image 8.2**. [2]

Image 8.2

Normal allele	C T T	T G T	A G T
Amino acid sequence	leu	-	
Mutant allele	C T T	T A T	A G T
Amino acid sequence	leu	-	

- (ii) Explain how the mutant allele will give rise to haemochromatosis. [3]

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- (c) (i) A woman, who is a carrier of the HFE mutation, and a man, who is affected by the condition are expecting a child.

Complete the Punnett square below to show the possible genotypes of their child. Use H to represent the normal allele.

[4]

Woman

Man

Phenotype ×

Genotype x

- (ii) State the probability of the couple having a child affected by haemochromatosis.

[1]

Probability =



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- (d) Genetic studies suggest the original mutant HFE allele arose in a single person who lived 60 to 70 generations ago. At that time, there was much less iron in people's diet.

Use the theory of natural selection to explain how the genetic condition haemochromatosis spread throughout the Northern European population at that time.

[4]

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TURN OVER FOR QUESTION 9

